



PERGAMON

www.elsevier.com/locate/watres

Wat. Res. Vol. 34, No. 1, pp. 117–126, 2000
© 1999 Elsevier Science Ltd. All rights reserved
Printed in Great Britain
0043-1354/99/\$ - see front matter

PII: S0043-1354(99)00097-4

FACTORS WHICH CONTROL BULK CHLORINE DECAY RATES

JAMES C. POWELL^{1*}, NICHOLAS B. HALLAM², JOHN R. WEST²,
CHRISTOPHER F. FORSTER² and JOHN SIMMS³

¹Binnie Black and Veatch Ltd, Grosvenor House, 69 London Road, Redhill, Surrey RH1 1LQ, UK;

²School of Civil Engineering, The University of Birmingham, Edgbaston, Birmingham B15 2TT, UK

and ³Severn Trent Water Ltd, Avon House, St Martins Road, Finham, Coventry CV3 6PR, UK

(First received 1 July 1998; accepted in revised form 1 February 1999)

Abstract—Several computer programs for modelling water distribution networks have been developed which incorporate a facility for modelling chlorine decay. Problems have been experienced with the calibration and durability of these models due to both temporal and spatial variability in the decay constants. Chlorine will decay either due to reactions at the pipe wall or due to reactions in the bulk water. The aim of the work presented in this paper is to investigate the factors which influence bulk decay. Over 200 determinations of bulk chlorine decay against time were performed on waters taken from 32 sampling locations within the Severn Trent region, U.K. The bulk decay constant was observed to show significant variation with temperature, the initial chlorine concentration and the organic content of the water. An equation was derived relating these parameters which could be used to update the decay constants in network models and improve their durability. © 1999 Elsevier Science Ltd. All rights reserved

Key words—bulk decay, chlorine, decay constant, network modelling

NOMENCLATURE

a	TOC coefficient (l/mg h)
b	temperature constant (°C)
C	chlorine concentration at time t (mg/l)
C_0	initial chlorine concentration (mg/l)
E	activation energy (J/mol)
e	slope of k_b against TOC (mg/l h)
F	frequency factor (units dependent on order of reaction)
f	TOC for zero k_b (mg/l)
g	slope of k_b against UV (cm/h)
H	concentration of reactant at time t (mg/l)
h	UV for zero k_b (l/cm)
m	slope of k_b against $1/C_0$ (mg/l h)
k	specific reaction rate constant (units dependent on order of reaction)
k	first order chlorine decay constant (l/h)
k_b	bulk first order chlorine decay constant (l/h)
k_b'	second order bulk decay constant (l/mg h)
k_w	wall first order chlorine decay constant (l/h)
p	universal k_b equation (TOC) coefficient (l/h)
q	universal k_b equation (UV) coefficient (mg cm/l h)
R	ideal gas constant (8.31 J/mol °C)
T	temperature (°C)
t	time (h)
TOC	total organic carbon concentration (mg/l)
UV	absorbance of ultraviolet light at a wavelength of 254 nm (l/cm)

INTRODUCTION

A number of network modelling packages incorpor-

ate a facility for modelling chlorine decay. Several studies have shown these models to be capable of modelling the chlorine concentrations observed at one moment in time (Chambers *et al.*, 1995; AWWARF, 1996). However, it is readily apparent that chlorine decay characteristics in a network show considerable variability with time. Most network modelling packages fail to account for any temporal variability in the decay characteristics and so their durability is restricted. These models generally assume that chlorine decays by first order kinetics (Chambers *et al.*, 1995; AWWARF, 1996; Stoner Associates, 1997):

$$C = C_0 \exp(-kt). \quad (1)$$

It is widely accepted that chlorine in distribution systems will either decay due to reactions with compounds contained within the bulk water or due to reactions at the pipe wall. This has led some researchers to model wall decay and bulk decay separately. A simple method for this is to define the overall decay constant as the sum of a bulk and wall decay constant (k_b and k_w):

$$k = k_b + k_w. \quad (2)$$

Investigating bulk decay is simpler than investigating *in situ* decay. Experiments can be performed under laboratory conditions with factors like temperature and reactant concentrations readily controlled. Bulk decay is normally measured by

*Author to whom all correspondence should be addressed.

recording the chlorine concentration, at time intervals, from glass bottles which have been previously filled with the sample water. The exact procedure varies, but it is often referred to as a "bottle" or "jar" test. Table 1 lists some first order decay constants obtained from literature and show that the value of k_b has been observed to vary between 0.02 and 0.74 l/h. This implies that there are factors affecting the decay rate which are not accounted for in the first order model.

The rate of virtually all chemical reactions increases with temperature. The Arrhenius equation is widely accepted (Coulson and Richardson, 1964):

$$k = F \exp[-E/R(T + 273)]. \quad (3)$$

The equation is normally given in terms of absolute temperature, but the Celsius scale is used here to maintain consistency of units. A very approximate rule stated by van't Hoff is that the reaction rates double for every 10°C rise in temperature (Fair and Geyer, 1954). This is equivalent to a value of 5750°C for E/R in the Arrhenius equation (Eq. (3)) for a temperature rise from 10–20°C.

Bulk chlorine decay rates have been observed to increase with temperature by several groups (Feben and Taras, 1951; Jadas-Hécart *et al.*, 1992; Zhang *et al.*, 1992) but none of this work has defined the relationship. The AWWARF (1996) reported an extensive study of the effect of temperature and total organic carbon concentration (TOC) on chlorine decay rates. This report states that unspecified previous work has shown that:

$$k = a\text{TOC} \exp[-b/(T + 273)]. \quad (4)$$

No data are shown in the report to verify this equation. It may be noted that b is effectively the ratio E/R in Eq. (3). They estimated a value of 1.8×10^6 l/mg h for a and a value of 6050°C for b . This would give a 2.05-fold rise in the decay constant between 10 and 20°C.

TOC is a widely used measure of organic content. The AWWARF (1996) reported a linear relationship between the first order decay constant and TOC (Eq. (4)), but no data were given. Some of the most detailed published work into bulk chlorine decay kinetics is that by Jadas-Hécart *et al.*, 1992. This work observed the decay of total chlorine in 15 water samples which had been dosed in the laboratory to an initial total chlorine concentration of

4 mg/l. The work differs from most other investigations of chlorine decay by using a two-phase second order model rather than the more common first order model. This model incorporates a parameter which should be related to the concentration of reactants, providing the stoichiometry of the reaction between chlorine and its reactants remained constant. However, they were unable to correlate this parameter with TOC. This may have been due to variations in the relative concentrations of organic compounds in the test samples.

Although the chlorine demand due to the pipe wall has been shown to decrease significantly with chlorine concentration (AWWARF, 1996), the authors have identified no studies relating specifically to the bulk demand. There also appears to have been relatively little work relating pH to chlorine decay. Given the extreme sensitivity of chlorine speciation to pH (AWWARF, 1992) this is, perhaps, surprising.

The aim of the work presented in this paper is to investigate the factors which affect bulk chlorine decay and to develop functional relationships that can be used to enhance the durability of network models. All the research relates exclusively to the decay of free chlorine. All the water samples were taken from distribution networks or water treatment works operated by Severn Trent Water Ltd.

MATERIALS AND METHODS

All chlorine measurements were taken by the DPD colourimetric method using Hach pocket chlorine colourimeters (pocket Hachs). It should be noted that, all spot samples were taken in duplicate and cross-checked on two pocket Hachs. The duplication of measurements reduced the error and proved a valuable quality control measure. Analysis involving the repeatability of measurements and a comparison with titration measurement techniques concluded that this procedure was capable of giving an accuracy of ± 0.02 mg/l (95% confidence level). All the pocket Hachs have been checked against two sets of standard proprietary coloured gels at least once a week when in use, but were always found to read within 0.02 mg/l of the supposed value.

All temperature readings were taken with standard mercury thermometers (accuracy of $\pm 1^\circ\text{C}$). All pH measurements were taken with a Hannah HI 9214 stick pH probe (accuracy ± 0.02 pH), which was regularly checked against standard buffer solutions). All UV measurements give the absorbance as measured by a Cecil CE 292 spectrophotometer using a wavelength of 254 nm with a 1 cm sample cell. All TOC measurements were performed by Severn Trent Laboratories using a Shimadzu 5000 automated analyser. TOC samples were collected in new bottles and analysed on the same day.

There is no standard test for measuring bulk chlorine decay, hence the following procedure was developed.

1. All glassware was treated by the following method (Department of the Environment, 1981) so that it was chlorine demand free. Freshly cleaned glassware was filled with distilled water which had been dosed to 10 mg/l free chlorine using concentrated sodium hypochlorite solution and left to stand for 24 h. It was then

Table 1. First order decay constants from literature

k_b (l/h)	Number of tests	Refs.
0.07–0.11	3	Zhang <i>et al.</i> (1992)
0.02	1	Clark <i>et al.</i> (1993)
0.03–0.21	2	Chambers <i>et al.</i> (1995)
0.01–0.74	10	AWWARF (1996)

Table 2. Sample sites for bulk decay tests. Amb: bulk tests measuring the decay of the ambient chlorine and Re Cl: bulk tests measuring the decay in a rechlorinated sample

Sample site	Comments	No. tests	
		Amb	Re Cl
Melbourne WTW final water	lowland river water; major treatment improvements including installation of GAC commissioned during survey	26	7
Melbourne SP05	part treated water entering super-chlorination		12
Melbourne SP12	part treated water leaving GAC and super-chlorination		9
Ab Kettleby reservoir	supplied from Melbourne WTW	15	
Haresfield network inlet	GAC treated lowland river water, rechlorinated twice	26	
Strensham WTW	GAC treated lowland river water	17	3
Birmingham University tap water	upland water		14
Others		35	44

emptied, rinsed thoroughly with distilled water and left to dry.

2. A 2.5-l Winchester bottle was filled with the sample water and left for 15 min to ensure homogeneity.
3. The sample water was decanted into eight 125-ml brown glass bottles and sealed with glass stoppers.
4. All bottles were stored in an incubator, ideally set at the temperature of the sample water. Samples taken in the field were stored on site in an incubator and transferred to the laboratory in an insulated box.
5. The chlorine concentration was measured at intervals from the bottles. The intervals were ideally set so that the chlorine concentration fell by 10% of the initial reading between measurements.
6. The decay constants were then calculated by a Visual Basic macro which optimised their value to minimise the sum of the squared errors between the modelled and observed chlorine concentrations.

The majority (60%) of the bulk decay tests measured the decay of the ambient chlorine in the initial water sample. The remaining tests were performed on samples in which the ambient chlorine concentration was too low to

allow accurate measurement of the decay. These samples were chlorinated with concentrated sodium hypochlorite solution prior to step 2 above giving initial chlorine concentrations between 0.5 and 2.5 mg/l. Some previous work has observed inconsistent behaviour in the period immediately after chlorination (Jadas-Hécart *et al.*, 1992). However, the samples from this study were generally not freshly chlorinated as they measured the decay of the ambient chlorine. Furthermore, all samples were left to rest for 15 min prior to taking the first chlorine measurement. Since most of the water in a distribution system is likely to be considerably more than 2 h old (since chlorination), this approach should give results which are more applicable to network modelling.

Four control bulk tests were performed on distilled water which had been dosed with sodium hypochlorite solution to give an initial chlorine concentration of approximately 0.60 mg/l. These tests consistently produced k_b values below 0.005 l/h, which is less than 10% of the average recorded value from the study. As a second check of the methodology, 21 of the bulk tests from the study were performed in duplicate. The standard deviation of the

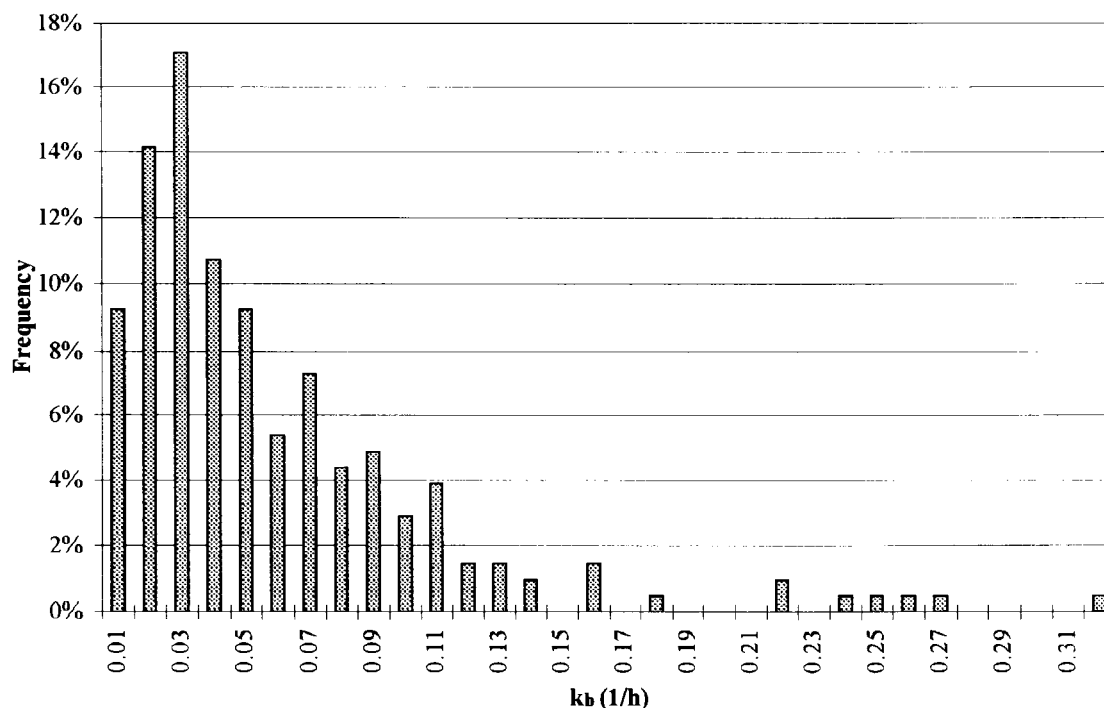
Fig. 1. Frequency distribution of k_b .

Table 3. Ranges of parameters measured

Sample site	Temperature (°C)		TOC (mg/l)		UV (l/cm)		C_0 (mg/l)		pH	
	No.	range	No.	range	No.	range	No.	range	No.	range
Melbourne WTW final water	26	4–20	26	0.9–4.2	1	0.024	26	0.21–0.56	7	6.8–7.3
Melbourne SP05	12	10–19					12	2.10–0.53	6	6.9–8.0
Melbourne SP12	9	5–18					9	0.24–2.50	9	6.8–7.5
Ab Kettleby reservoir	15	8–23					15	0.12–0.32	4	6.9–7.4
Haresfield network inlet	26	8–22	6	1.4–2.9	26	0.025–0.046	26	0.16–0.99	20	6.9–8.3
Strensham WTW	17	4–15					17	0.31–0.71	12	6.8–7.3
Birmingham University tap water	14	14–26			2	0.035–0.036	14	0.70–0.50	6	9–9.1

difference between replicate values was ± 0.010 l/h, which when analysed assuming a student's *t*-distribution, gave a standard error of ± 0.011 l/h and 95% confidence limits of ± 0.022 l/h.

Temperature effects were investigated in the laboratory on water samples collected from three different sources. For each water sample, bulk tests were undertaken simultaneously at three different temperatures, with the initial chlorine concentrations dosed to approximately 0.5 mg/l.

Between June 1996 and September 1997, 207 determinations of bulk chlorine decay against time were performed, on waters taken from 32 sampling locations. Table 2 lists the primary water sources. The frequency distribution of k_b values obtained in this program of tests is shown in Fig. 1. 90% of k_b values are between 0.008 and 0.16 l/h. This is a wide range, since water with a k_b value of 0.16 l/h will decay to a minimum acceptable level in one 20th of the time of water with a k_b value of 0.08 l/h. In an attempt to understand this variability, the k_b value has been analysed with respect to temperature, TOC, UV, C_0 and pH. The ranges of these parameters for the different water sources surveyed is shown in Table 3.

RESULTS

The results of the laboratory investigation of the effect of temperature on k_b are shown in Fig. 2. The Arrhenius equation was fitted to each set of tests by optimising the values of F and E/R to minimise the sum of the squared errors between predicted and observed k_b values. A good fit between modelled and observed data was achieved giving R^2 fits in excess of 0.98. The maximum deviation between modelled and observed decay rates was 0.004 l/h which is well within the measurement errors estimated at ± 0.01 l/h.

For the field data, a significant relationship was found at three sites. The R^2 correlation coefficients and the optimised values for F and E/R for the field and the laboratory tests are shown in Table 4. When both F and E/R are optimised, the values of E/R range between 4660 and 9560°C. This results in a 1.8- to 3.2-fold rise in decay rates between 10 and

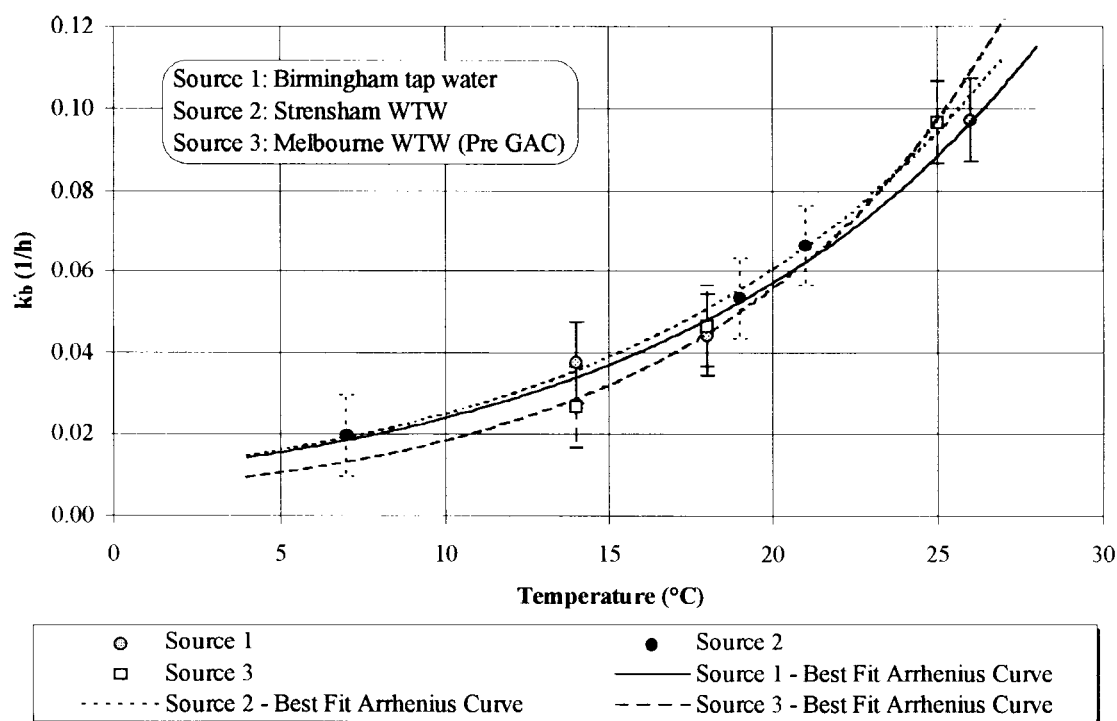
Fig. 2. k_b vs temperature for laboratory tests.

Table 4. Best fit Arrhenius coefficients

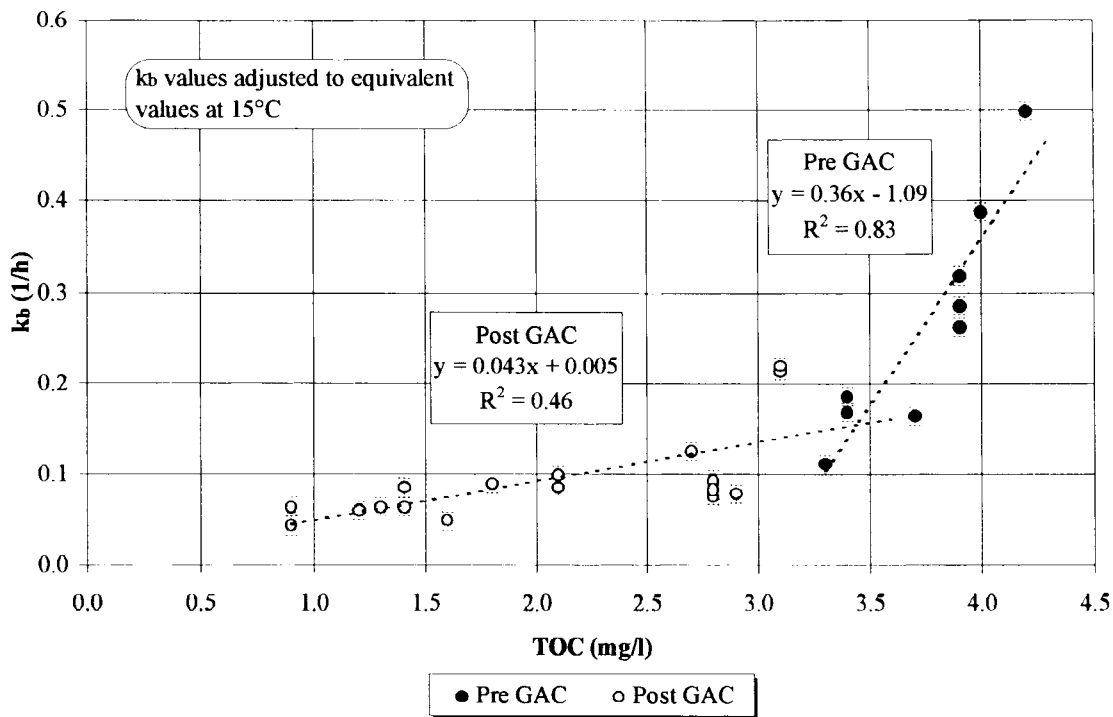
Test	Details of best fit Arrhenius curves							
	<i>n</i>	<i>F</i> and <i>E/R</i> optimised			<i>E/R</i> = 6000°C		<i>E/R</i> = 7300°C	
		<i>F</i> (1/h)	<i>E/R</i> (°C)	<i>R</i> ²	<i>F</i> (1/h)	<i>R</i> ²	<i>F</i> (1/h)	<i>R</i> ²
Lab test Birmingham tap water	3	8.13 × 10 ⁹	7,520	0.99	4.74 × 10 ⁷	0.98	3.84 × 10 ⁹	0.98
Lab test Strensham WTW	3	4.03 × 10 ⁹	7,300	1.00	4.63 × 10 ⁷	0.99	3.96 × 10 ⁹	1.00
Lab test Melbourne WTW (pre GAC)	3	8.50 × 10 ¹²	9,560	1.00	4.79 × 10 ⁷	1.00	3.96 × 10 ⁹	1.00
Field data Strensham WTW	17	1.80 × 10 ⁸	6,380	0.69	4.77 × 10 ⁷	0.68	4.60 × 10 ⁹	0.69
Field data Haresfield network inlet	26	7.82 × 10 ⁵	4,660	0.29	4.84 × 10 ⁷	0.28	4.19 × 10 ⁹	0.27
Field data Melbourne WTW (post GAC)	15	3.91 × 10 ¹¹	8,410	0.85	8.90 × 10 ⁷	0.84	8.19 × 10 ⁹	0.84
Average		1.48 × 10 ¹²	7,300	0.80	5.45 × 10 ⁷	0.80	4.79 × 10 ⁹	0.80

20°C. This variation in E/R is accompanied by a 10^7 -fold variation in F , suggesting that the parameter is unstable. This would make it difficult to apply in modelling scenarios. Fixing E/R at a constant value greatly increases the stability of F reducing the 10^7 -fold variability to a two fold variation. Using a value of 6000°C for E/R , as found by the AWWARF (1996), produces a curve that falls within the majority of the error bands, but tends to underestimate the gradient. This is overcome by using the average optimised value of E/R which was 7300°C. This produces a 2.5-fold increase in decay rates between 10 and 20°C compared to a twofold rise for an E/R value of 6000°C.

The strong relationship with temperature can mask more subtle relationships with other parameters. The Arrhenius relationship is used in subsequent analysis to calculate equivalent decay

coefficients at 15°C which can be directly compared. This was done by assuming a value of 7300°C for E/R and calculating the value of F for each test using Eq. (3). The same equation was then used to calculate what k_b value would be at a different temperature assuming that F and E/R remained constant. The same technique could be used to update the decay coefficients in chlorine network models when the temperature changes from that experienced when the model was calibrated.

The final water from Melbourne WTW was the only source from the data set where a comprehensive record of TOC was recorded. Across the sample period, it exhibited a large variation (0.9–4.2 mg/l) due to the introduction of granular activated carbon (GAC) treatment. The k_b values adjusted to equivalent values at 15°C are plotted against TOC in Fig. 3. A strong positive relation-

Fig. 3. Melbourne WTW — k_b vs TOC.

ship was observed between TOC and k_b , but there is a sharp change in gradient after the installation of GAC. The change in gradient could be due to a change in the chemical matrix of the water following the installation of GAC. Whilst TOC should give some measure of the quantity of organics available for chlorine to react with, it does not give any measure of the relative reactivity of the constituents of this organic matrix.

An alternative measure of organic content is UV. A relationship was observed between UV and k_b values (adjusted to equivalent values at 15°C) collected over a four-week period in summer 1997 at the inlet to the Haresfield network (Fig. 4). It is encouraging that a relationship was observed despite a relatively small range in UV values (0.025–0.046 l/cm). UV tends to give a measure of unsaturated organic bonds. These are potentially sites with which chlorine can react so, unlike TOC measurements, it is a measure of both the quantity and the stoichiometry of the organics in the water.

Previous research had proposed that decay rates were directly proportional to TOC (Eq. (4)). However, none of the TOC or UV relationships observed in this study pass through the origin. Hence equations of the following form would be required to model them:

$$k_b = e(\text{TOC} - f) \quad (5)$$

$$k_b = g(\text{UV} - h). \quad (6)$$

The optimal value of e was found to be 0.36 mg/l h for samples from Melbourne WTW prior to the installation of GAC decreasing to 0.04 mg/l h after the installation of GAC. The optimal value of f was found to be 0 mg/l for samples from Melbourne WTW prior to the installation of GAC and −0.12 mg/l after the installation of GAC. The optimal values for g and h were found to be 1.61 cm/h and 0.012 l/cm, respectively, for samples from the Haresfield network Inlet.

It became readily apparent that if two identical samples were chlorinated to different initial chlorine concentrations (C_0), the sample with the higher dose produced a lower decay rate. The data set was sorted for all tests which were done in duplicate, but with a difference of greater than 25% between the C_0 of the replicates. This found 16 pairs of tests from a wide variety of sources. In each case, k_b was lowest in the sample with the highest C_0 . Plotting the product of k_b and C_0 for each test against that of its pair produced a near straight line passing close to the origin ($r^2=0.70$ increasing to 0.90 if one pair is excluded). This implies that k_b is inversely proportional to C_0

$$k_b = m/C_0. \quad (7)$$

Analysis of samples which were not taken simultaneously is prone to noise due to variations in the temperature and chemical matrix of the water. Nevertheless, relationships were observed between k_b and C_0 at several sites. The relationship observed

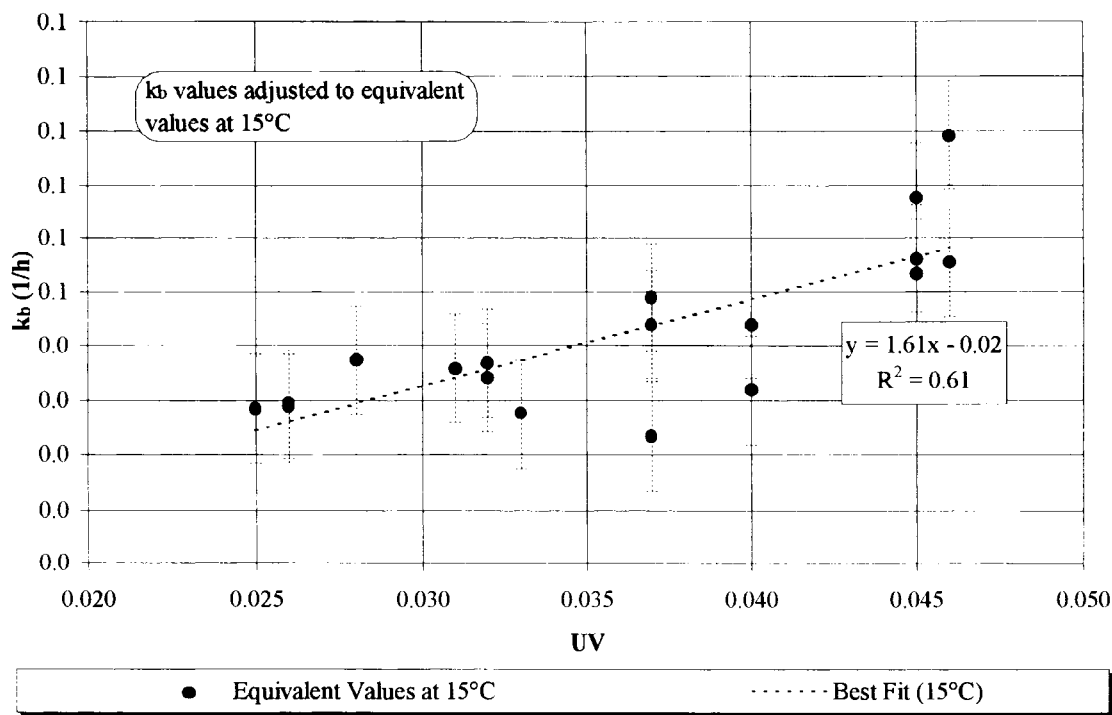


Fig. 4. k_b vs UV for samples from the Haresfield Network inlet.

at the Haresfield network Inlet during a four-week period in spring 1997 is shown in Fig. 5.

No significant relationship was found between k_b and pH at any sample point. This was probably due to the small range of pH values observed.

Combining the relationships observed between k_b and temperature, organic content and C_0 yields the following equations: With respect to TOC,

$$k_b = p(1/C_0)(\text{TOC} - f) \exp[-E/R(T + 273)]; \quad (8)$$

or with respect to UV,

$$k_b = q(1/C_0)(\text{UV} - h) \exp[-E/R(T + 273)]. \quad (9)$$

Eq. (8) was applied to the data collected from Melbourne WTW. The value of E/R was set at 7300°C, the average optimal value from the analysis of temperature effects (Table 4). The constants p and f were derived by optimisation to minimise the sum of the squared errors between predicted and observed k_b values. Due to the significantly different behaviour between the water pre- and postinstallation of GAC, separate constants were derived for each of these data sets. The modelled and observed k_b values are plotted in Fig. 6. The maximum error is 0.08 l/h or 80%, however 60% of the modelled data is within 0.02 l/h or 20% of the observed data. There is a tendency for the errors to increase at higher decay rates. This might be due to a higher organic content, in which the chemical matrix of the water and the stoichiometry of the reactants becomes increasingly significant.

Eq. (9) was applied to the full data set from the inlet to the Haresfield network. Again a value of 7300°C was used for E/R and the parameters q and h were optimised to minimise the sum of the squared errors between predicted and observed decay rates. Fig. 7 shows that 60% of the modelled data lie within the measurement errors of the observed data (± 0.01 l/h) and 85% are within 0.02 l/h. Despite the errors being generally lower than in the Melbourne data, the correlation is weaker with an R^2 of 0.60 compared to 0.85. This is due to the k_b values at the Haresfield network being on average one third of those at Melbourne.

It was observed that if a sample of water was allowed to decay and then rechlorinated to a concentration similar to the original concentration, the decay rate of the rechlorinated sample was significantly less than that of the original sample. This exercise was repeated six times with water from various sources and in every case the decay rate of the rechlorinated sample was reduced by at least 40% (Fig. 8). It is possible that the reduction in decay rates was caused by a reduction in the concentration of available organic reactants. This effect suggests that the common practice of super-chlorination will have a significant impact on the decay rate, and any changes to the operating procedures will affect the decay. Also, the use of chlorine booster stations within distribution systems to rechlorinate water will reduce k_b and allow chlorine to penetrate further into the system than might be anticipated.

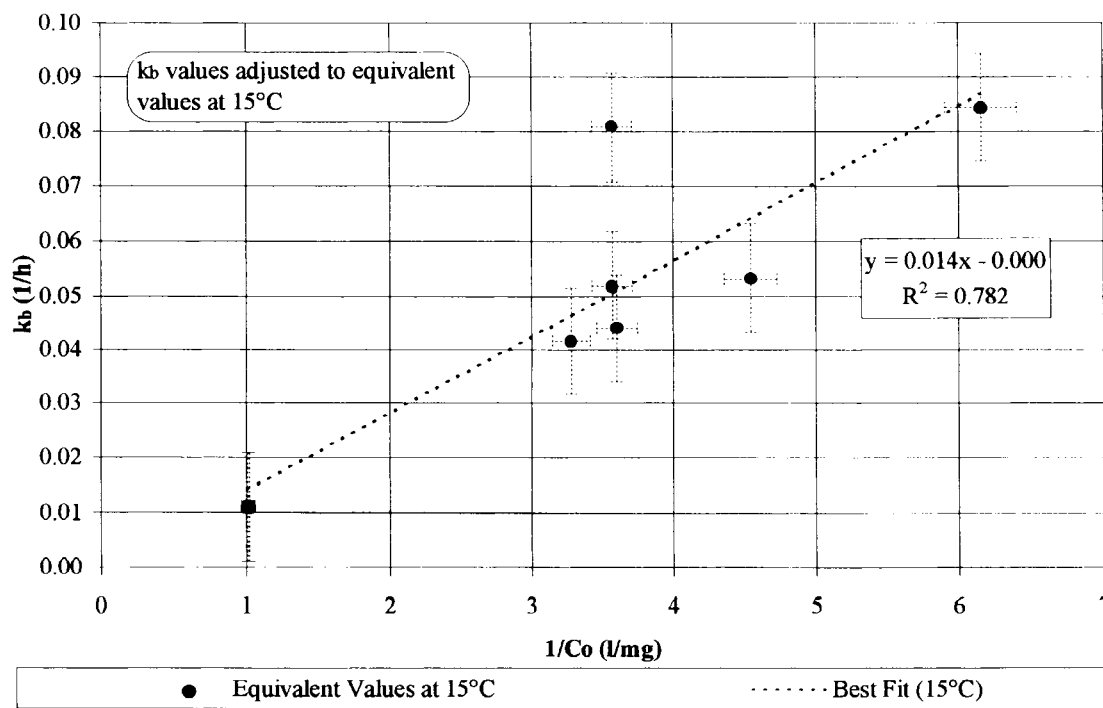
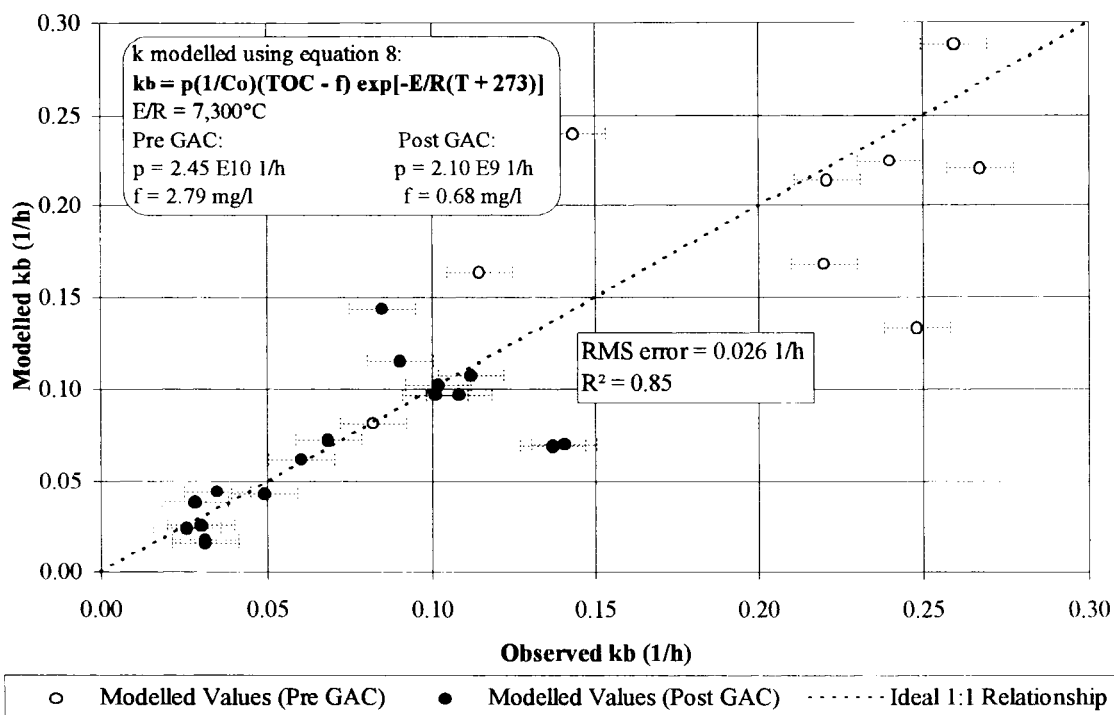


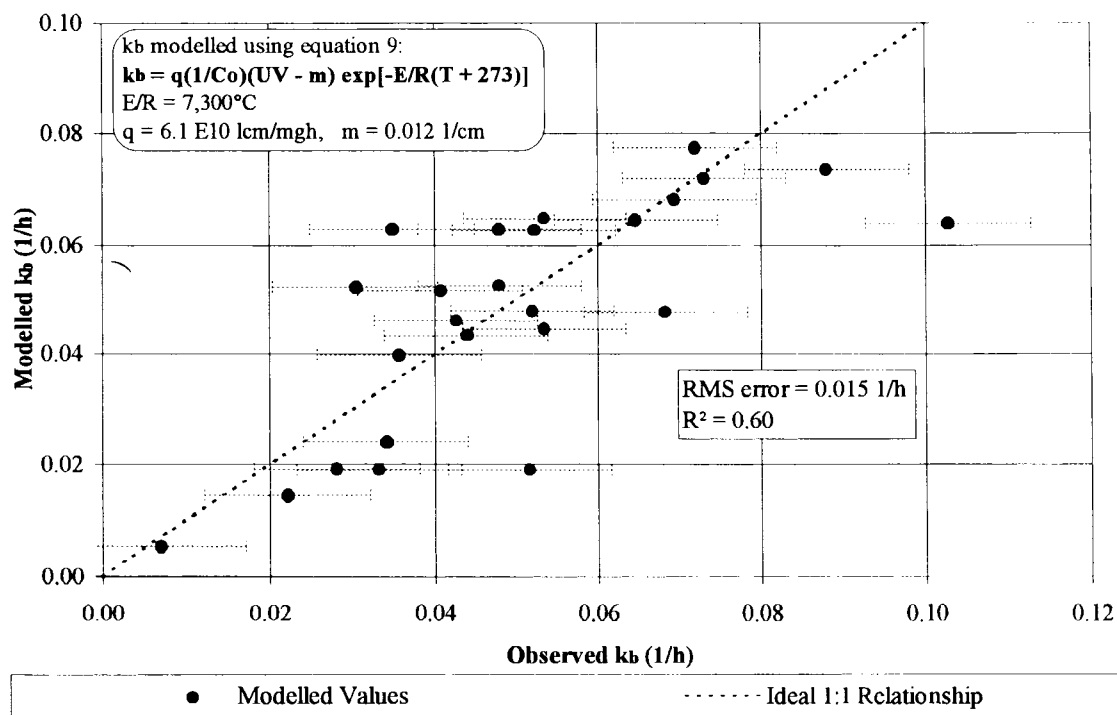
Fig. 5. k_b vs $1/C_0$ for samples from the Haresfield Network inlet.

Fig. 6. k_b values modelled using Eq. (8) vs measured values at Melbourne WTW.

DISCUSSION

Although reaction rate constants can vary with temperature or other solution conditions, they should be independent of the concentration of the

reacting compounds (Fair and Geyer, 1954). Despite the reasonable fit of the decay curves to a first order profile, three of the observations in this paper contradict this. First that k_b is inversely pro-

Fig. 7. k_b values modelled using Eq. (9) vs measured values at the Haresfield Network inlet.

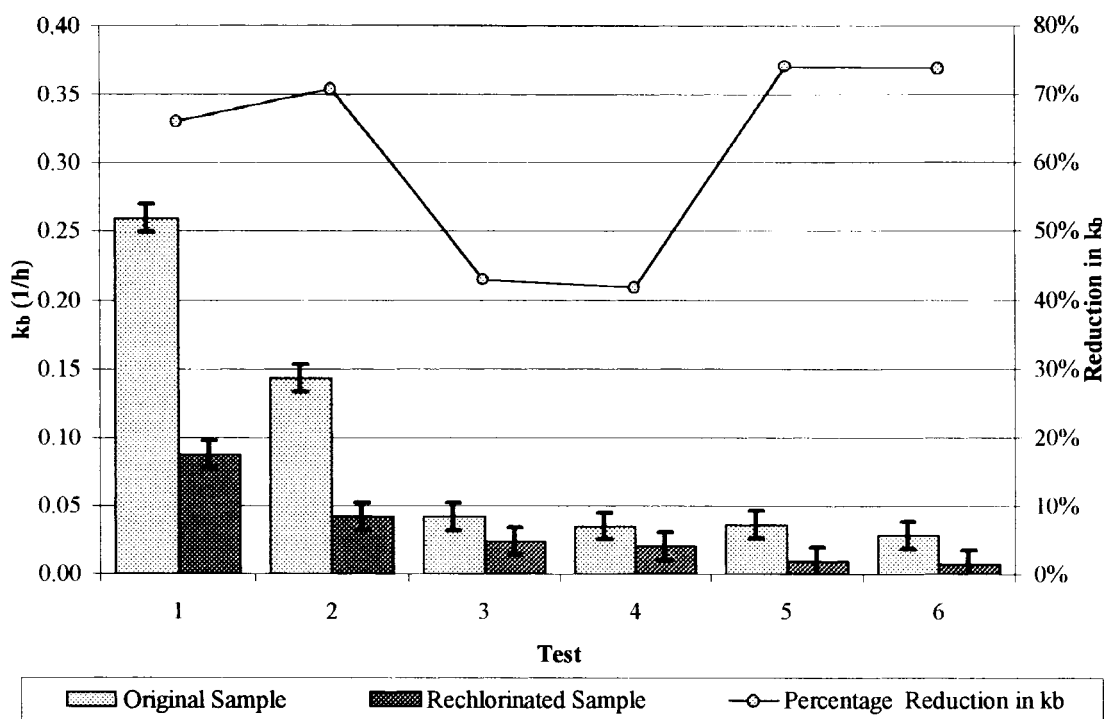


Fig. 8. Effect of rechlorination on k_b .

portional to C_0 , second that k_b is reduced by rechlorination and third that k_b is proportional to measures of the organic concentration (TOC or UV).

A possible partial explanation of the above is that the first order algorithm is being used to model what is actually a second order decay process. Second order decay assumes

$$dC/dt = -k_b CH. \quad (10)$$

TOC and UV are likely to be indications of the concentration (H) of the compounds with which chlorine reacts. A second order mechanism would explain why the decay rate was observed to increase with TOC or UV. The concentration of the available reactants will reduce along with the chlorine concentration. Hence, H will be lower in a rechlorinated sample than in the original sample. This would result in a lower rate of decay in the rechlorinated sample. The higher the initial chlorine concentration, the greater the concentration of the reactants that will probably be utilised in the decay process. Hence the average concentration of reactants across the decay period will be lower. This in turn would reduce the average decay rate across the decay period.

Where the concentration of the reactants (H) is in large excess, the concentration will not change much during the course of the reaction and will be effectively constant. It is clear from Eq. (10) that if H is effectively constant then the second order

decay will approximate a first order decay. This is often referred to as "pseudo first order" (Fair and Geyer, 1954). The average value of C_0 in this study was 0.5 mg/l. It is worth noting that this is much lower than that in some previous research. The average value of C_0 in the tests by the AWWARF (1996) was 1.0 mg/l and the C_0 values of the work by Jadas-Hécart *et al.* (1992) and Dossier-Berne *et al.* (1997), investigating the use of second order kinetics, were between 4 and 15 mg/l. At these higher doses, much more of the organic content is likely to react and so it is less likely that the decay will be pseudo first order. However, it has been shown for the observed waters, that a first order model can give a reasonable fit to the shape of the decay profile (Powell, 1998). Until the concentration of the reactants (H) and the stoichiometry can be readily measured, a second order decay model, taking into account the reactant concentration will not be a pragmatic option for network modelling of good quality waters.

CONCLUSIONS

1. k_b was observed to vary with temperature, measures of the organic content (TOC and UV) and the initial chlorine concentration (C_0). An equation was derived relating these parameters to k_b . The important elements of this equation are an Arrhenius type relationship with temperature (optimum value of $E/R = 7300^\circ\text{C}$ producing a

- 2.5-fold increase in decay rates between 10 and 20°C); a linear increase with TOC or UV and an inverse relationship with C_0 .
2. No relationship was observed between pH and k_b , possibly due to the small pH range observed.
3. In all observed cases rechlorination of a sample reduced the value of k_b .
4. Some of the observed behaviour contradicts the assumption of first order kinetics. Second order kinetics can explain this behaviour partially. However, for viable network modelling, all the parameters in the decay algorithm utilised must be readily quantified. At present this would be difficult for a second order algorithm. The most pragmatic method is to assume first order decay, but to define the decay constant as a function of temperature, measures of the organic content (TOC and UV) and the initial chlorine concentration (C_0).

REFERENCES

- AWWARF (1992) *Disinfectant Residual Measurement Methods*, 2nd edn. AWWA, USA.
- AWWARF (1996) *Characterisation and Modeling of Chlorine Decay in Distribution Systems*. AWWA, USA.
- Chambers V. K., Creasy J. D. and Joy J. S. (1995) Modelling free and total chlorine decay in potable water distribution systems. *J. Water Sci. Res. Technol.: Aqua* **44**, 60–69.
- Clark R. M., Grayman W. M., Males R. M. and Hess A. F. (1993) Modeling contaminant propagation in drinking-water distribution systems. *J. Environ. Eng.* **119**, 349–364.
- Coulson J. M. and Richardson J. F. (1964) *Chemical Engineering*, 2nd edn. Pergamon Press, Oxford, (3 volumes).
- Department of the Environment (1981) *Methods for the Examination of Waters and Associated Materials: Chemical Disinfecting Agents in Water and Effluents and Chlorine Demand*. HMSO, London.
- Dossier-Berne F., Panais B., Merlet N., Jadas-Hécart A., Cauchi B. and Legube B. (1997) Automation of long term chlorine demand measurement of treated waters. *Water Res.* **31**, 375–384.
- Fair G. M. and Geyer J. C. (1954) *Water Supply and Waste-water Disposal*. John Wiley and Sons, London.
- Feben D. and Taras M. J. (1951) Studies on chlorine demand constants. *J. Am. Water Works Assoc.* **43**, 922–932.
- Jadas-Hécart A., El Moher A., Stitou M., Bouillot P. and Legube B. (1992) The chlorine demand of a treated water. *Water Res.* **26**, 1073–1084.
- Powell J. C. (1998) Modelling chlorine in water distribution networks. Ph.D. thesis, The University of Birmingham.
- Stoner Associates (1997) *Stoner Workstation Service for Water Users Manual. Version 2.3*. Stoner Associates, Houston, TX.
- Zhang G. R., Kiene L., Wable O., Chan U. S. and Duguet J. P. (1992) Modelling of chlorine residual in the water distribution network of Macao. *Environ. Technol.* **13**(10), 937–946.